COMP 561
Assignment 2
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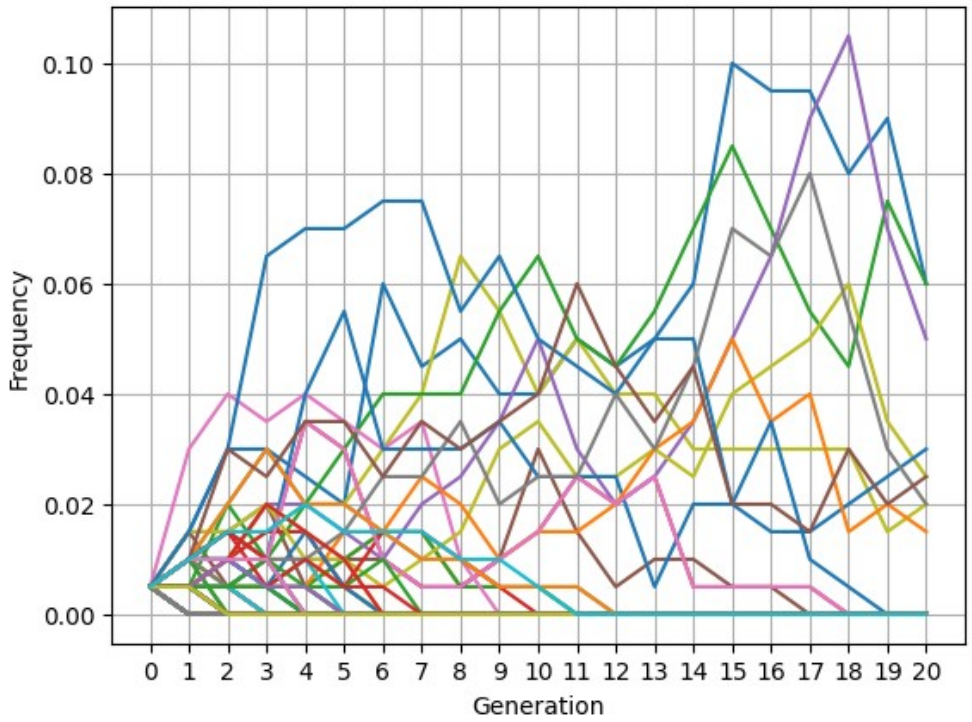
Question 1.

a) The code for question 1 can be found in the submitted file named `q1.ipynb`.

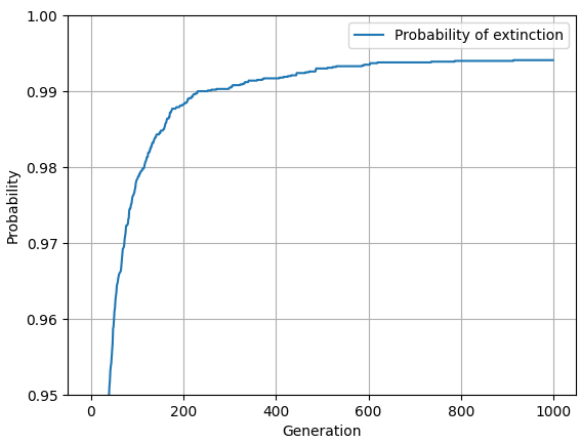
b) The alternative allele becomes extinct at generation 1 if none of the 2N copies inherit it. The probability of drawing the reference allele is 1 – (1/(2N). The probability that all 2N copies are from the reference allele is $\left(1-\frac{1}{2N}\right)^{2N} = \left(1-\frac{1}{200}\right)^{200} = 0.36696$ meaning 36.696%

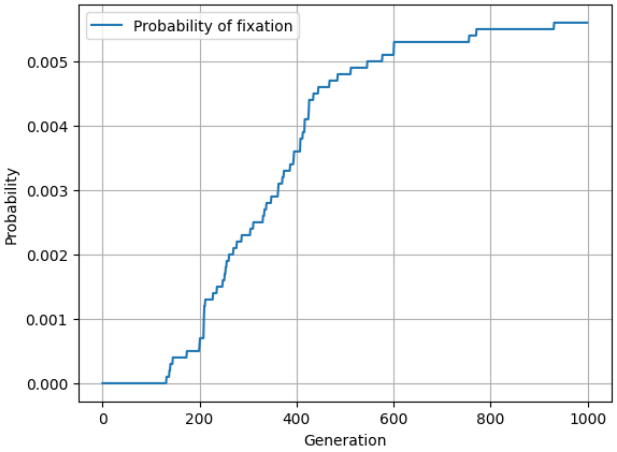
c) We find that 36.7634% of SNPs go to extinction at generation 1 which closely matches our analytical result.

d) Plotting the evolution of allele frequency of the first 100 SNPs across 20 generations, I get a figure quite similar to the one in the handout. Most alleles go extinct, and none has enough time to fixate.



e) We see that 99.41% of alleles go extinct and 0.56% get fixated (0.03% are in between). These results approximately match the theoretical fixation and extinction rate of neutral alleles that is equal to its initial frequency (1/(2N) = 1/200 = 0.05). The rate of fixation is the complement of that (1-1/(2N) = 1-0.05 = 0.95).





f) This is quite similar to part b. Except that the probability of drawing the alternate allele for SNP42 is raised from 1/2N to 1.5/(2N+0.5). So the probability of it going instinct after 1 generation is:

$$\left(1-\frac{1.5}{200.5}\right)^{200}=0.2227=22.27\%$$

With my simulation, I get a probability of 22.24% which is close to the theoretical value.

g) Running the simulation, I get an extinction probability of 97.1% and a fixation probability of 0.1%.

h) Running the simulation, I get an extinction probability of 99.27% and a fixation probability of 0.14%.

Without any advantage, after 100 generations, I get an extinction probabiliy of 98.9% and a fixation probability of 0.246%

Question 2.

a) The program and following results can be found in the jupyter notebook file `q2.ipynb`.

b) I get 454 SNPs with uncorrected pvalue below (not including) 0.05. By chance, we expect $0.05 \times 10000 = 500$ SNPs to get a pvalue below 0.05 even when there is no association. That is larger but somewhat close to what we found.

c) I find 4 SNPs that have a corrected pvalue below 0.05 shown in the table below. Note that the SNP index starts from 0 (not 1). So the 4 SNP Ids are SNP1000, SNP2000, SNP3000, and SNP4000.

SNP Index	Uncorrected p-value	Corrected p-value	Odds ratio het	Odds ratio homo alt
1000	1.732482e-16	1.732482e-12	1.582846	2.509911
2000	9.230119e-07	9.230119e-03	1.503494	2.248627
3000	5.629213e-10	5.629213e-06	1.587617	2.027660
4000	4.770372e-07	4.770372e-03	1.344102	1.996176

d) This is expected and can occur due to a multiple of reasons. The number of observations in each element of the contingency table (for different genotypes and disease conditions) are not equal across SNPs. For example SP1 can have 100 healthy samples who are heterozygous while SNP2 can have 500. Having a low allele frequency can change the statistical power of the chi-squared test for the worse.

Question 3.

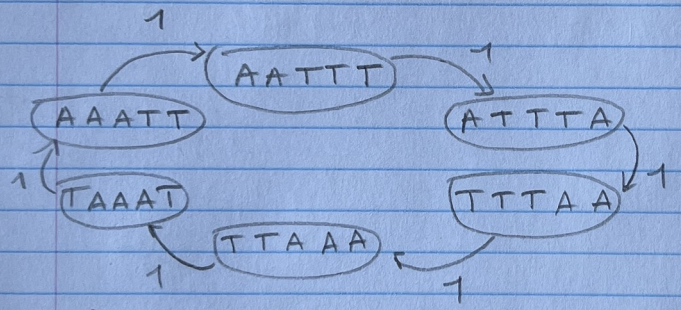
Consider the following sequence G

G = AATTTAAATT

with shotgun sequencing, we might get the following reads:

AATTT	TTAAA
ATTTA	TAAAT
TTTAA	AAATT

Attempting to form back the sequence from reads



if we start from the node AATTT,

G' = AATTTAAATT (original sequence)

but starting from another node say AAATT,

G'' = AAATTTAAAT

or from TAAAT,

G''' = TAAATTTAAA

Starting Assembly at each node results in a different, but equally scored answer.

Question 4.

a) The default parameters for blastn did not result in any matches. I'm using the standard databases (nr etc.) and the core nucleotide database (core nt). After tweaking with parameters, I got hits with good Evalues pictured below corresponding to a smaller word size of 7. The matches seem to correspond to Zika Virus that causes diseases including Zika fever.

select all *100 sequences selected* GenBank Graphics Distance tree of results MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Zika virus strain ZIKV/Homo sapiens/COL/FLR_00024/2015 polyprotein (GP1) gene, complete cds	Zika virus	127	127	100%	3e-25	83.33%	10659	MF574569.1
Zika virus isolate AB4537 polyprotein (POLY) gene, complete cds	Zika virus	127	127	100%	3e-25	83.33%	10602	OR264647.1
Zika virus isolate AB4070 polyprotein (POLY) gene, complete cds	Zika virus	127	127	100%	3e-25	83.33%	10596	OR264643.1
Zika virus isolate PR/DB-ZIKV266/2016 polyprotein (POLY) gene, complete cds	Zika virus	127	127	100%	3e-25	83.33%	10272	MW122388.1
Zika virus isolate AB4506 polyprotein (POLY) gene, complete cds	Zika virus	127	127	100%	3e-25	83.33%	10602	OR264646.1
Zika virus strain ZIKV/Homo sapiens/COL/FLR_00025/2015 polyprotein (GP1) gene, complete cds	Zika virus	127	127	100%	3e-25	83.33%	10659	MF574563.1
Zika virus strain ZIKV/Homo sapiens/COL/FLR_00026/2015 polyprotein (GP1) gene, complete cds	Zika virus	127	127	100%	3e-25	83.33%	10659	MF574564.1
Zika virus isolate ZIKV/Homo_Sapiens/NIC/7193_CHB3146/2016 polyprotein (POLY) gene, complete cds	Zika virus	123	123	100%	1e-23	82.50%	10769	OK054410.1
Zika virus strain Guadeloupe_Aedes_aegypti_T961 polyprotein gene, complete cds	Zika virus	123	123	100%	1e-23	82.50%	10585	MN185331.1
Zika virus isolate TRDP274 polyprotein gene, partial cds	Zika virus	123	123	100%	1e-23	82.50%	9464	OQ727572.1
Zika virus isolate ZIKV/Homo_Sapiens/NIC/5406_CZB146/2016 polyprotein (POLY) gene, complete cds	Zika virus	123	123	100%	1e-23	82.50%	10782	OK054429.1
Zika virus isolate PR/DB-ZIKV451/2016 polyprotein (POLY) gene, complete cds	Zika virus	123	123	100%	1e-23	82.50%	10272	MW122400.1
Zika virus isolate PR/DB-ZIKV540/2016 polyprotein (POLY) gene, complete cds	Zika virus	123	123	100%	1e-23	82.50%	10272	MW122408.1

Figure x – Results of running nblast with word size of 7

Prior to getting these results, I was getting high Evalue matches to walking sticks and a fungus (pictured below). Given that none of those were pathogens and had bad Evalues, I ignore them.

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
2_Tsi_b3v08	Timema shepardi	43.6	43.6	27%	5.0	90.91%	343848	OC000277.1
Glarea lozoyensis ATCC 20868 P-loop containing nucleoside triphosphate hydrolase (GLAREA_10791), pa...	Glarea lozoyens...	43.6	43.6	19%	5.0	100.00%	5688	XM_008079891.1
2_Tcm_b3v08	Timema californi...	43.6	43.6	27%	5.0	90.91%	96334	OE181698.1
4_Tte_b3v08	Timema tahoe	43.6	43.6	19%	5.0	100.00%	193909	OE001166.1

Figure x – Results of running megablast with word size of 16 (no correct hits)

b) Here, I assume that the treatment corresponding to the strain of the pathogen that is the closest to our mystery strain is more likely to be effective that one corresponding to an evolutionary distant strain. Running the LIRMM phylogenetic inference package results in the figure bellow with branch lenghts showing distances between strains (divergence of their sequences). The strain with the closest distance to our mystery strain is strain 4. The next closest is strain 5. So the treatment for strain 4 is the most likely to work following the described assumption.

D(mystery, s4) = 0.05709+0.06961+0.04242 = 0.16912
D(mystery, s5) = 0.05709+0.0.06652+0.0.07822 = 0.20183

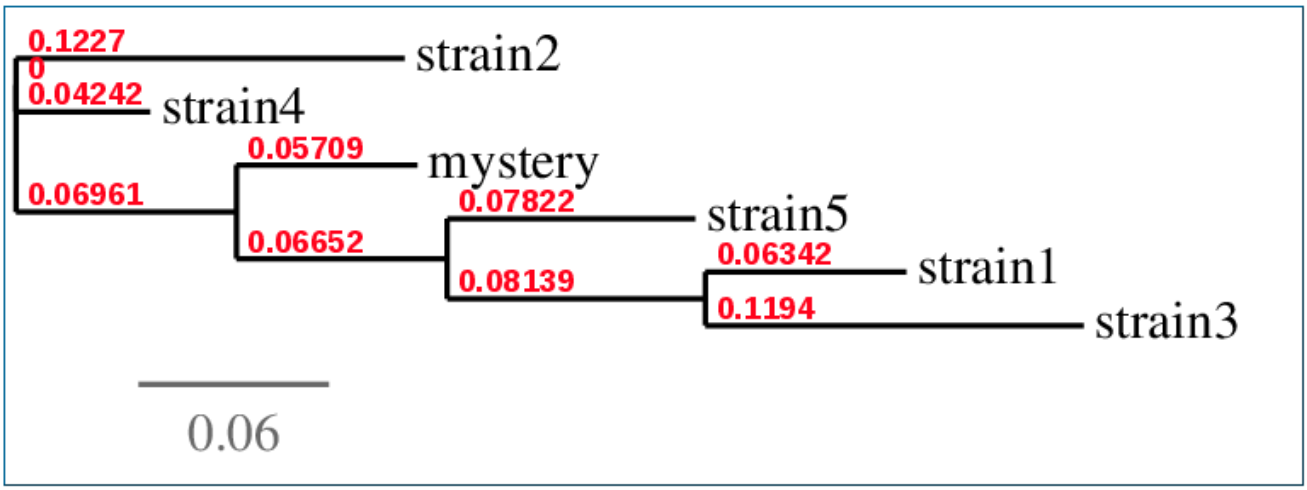


Figure 1: *Phylogenetic tree (the branch length is proportional to the number of substitutions per site).*

Question 5.
a)

Inputs: Tree T
Di: list of trait vectors for leaf nodes $\mathcal{U}$
M: max trait value (integer)
K: number of traits

// we first initialize a cost matrix C of dimension $\mathcal{U} \times K \times M$
// where C[u][k][m] denotes the minimal possible $\sum_{k=1}^{K} |D_{ui}(k) - D_v(k)|$ where the k_{th} trait has a value m at node u

C[u][k][m] = ∞ for k in 0...K-1 and m in 0...M

// Do a postorder traversal (from leaves to root)

For node u in PostOrder(T): // returns u in postorder
  if u is a leaf node:
    for k in 0...K-1:
      for m in 0...M:
        if Di[u][k] = m:
          C[u][k][m] = 0
  else: // if u is not leaf node
    v, w = children of (u, T) // given T and node, returns 2 children
    For k in K:
      For m in M:
        min_v = min(C[v][k][m] + abs(m-z) for z in 0...M)
        min_w = min(C[w][k][m] + abs(m-z) for z in 0...M)
        C[u][k][m] = min_w + min_v

// initialize empty D matrix to fill up during traceback
// (not to be confused with Di: leaf node trait vectors)
D = zeroes((|$\mathcal{U}$|-1, K)) // size internal nodes x k

// now do a preorder traversal (from root to leaves)

```
root = getRoot(T)
traceback(root, T)
output D, sum( C[root][k][D[root][k]] for k in 0...J-1 )
    // output the total final score and ancestral trait vectors
```

defined here

```
* Function traceback(node, Tree, parent_traits = None):
    if node is a leaf:
        return Di[node] // we have these as input already
    For k in 0...J-1:
      if node is the root:
        D[node] = argmin( C[node][k][m] for m in 0...M )
      else: // if node is not root
        pt = parent_trait(k)
        D[node] = argmin (C[node][k][m] + abs(m - pt) for m in 0...M

    //recurse
    v, w = children of(node, T)
    traceback(v, Tree, D[node])
    traceback(w, Tree, D[node])
```

b) The running time is

$$O((n-1).k.M^2)=O(n.k.M^2)$$

Initialization takes O(n.k.M).
during postorder and preorder traversals, we iterate over M possibilities for the two children and repeat that for k traits and n-1 nodes O((n-1).k.M^2). So it scales linearly with number of nodes and traits but quadratically with maximum trait value.

It should be noted that this approach is quite problematic as biological traits do not all have the same scale (age can be from 0-100 while lenght of forearm can be from 20-50 cm). That leads to mant parameters that are computed in our matrices but do not happen and won't get chosen. It might be helpful to somewhat normalize values and scale them accordingly. Or we could have a array of different lenght for traits with different M (example: size 100 for age, size 50 for lenght of forearm).